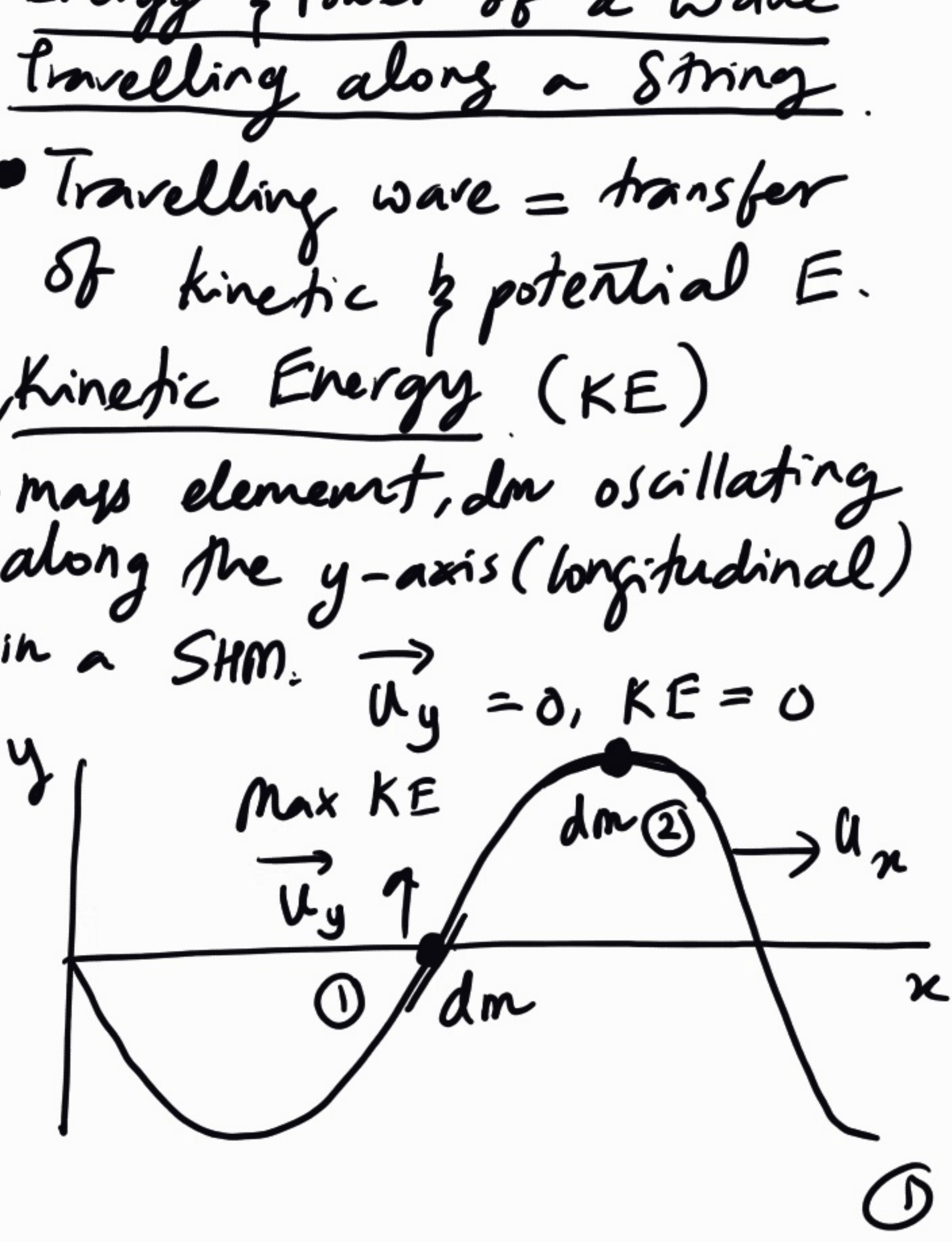
Energy & Power of a Wave Travelling along a String.

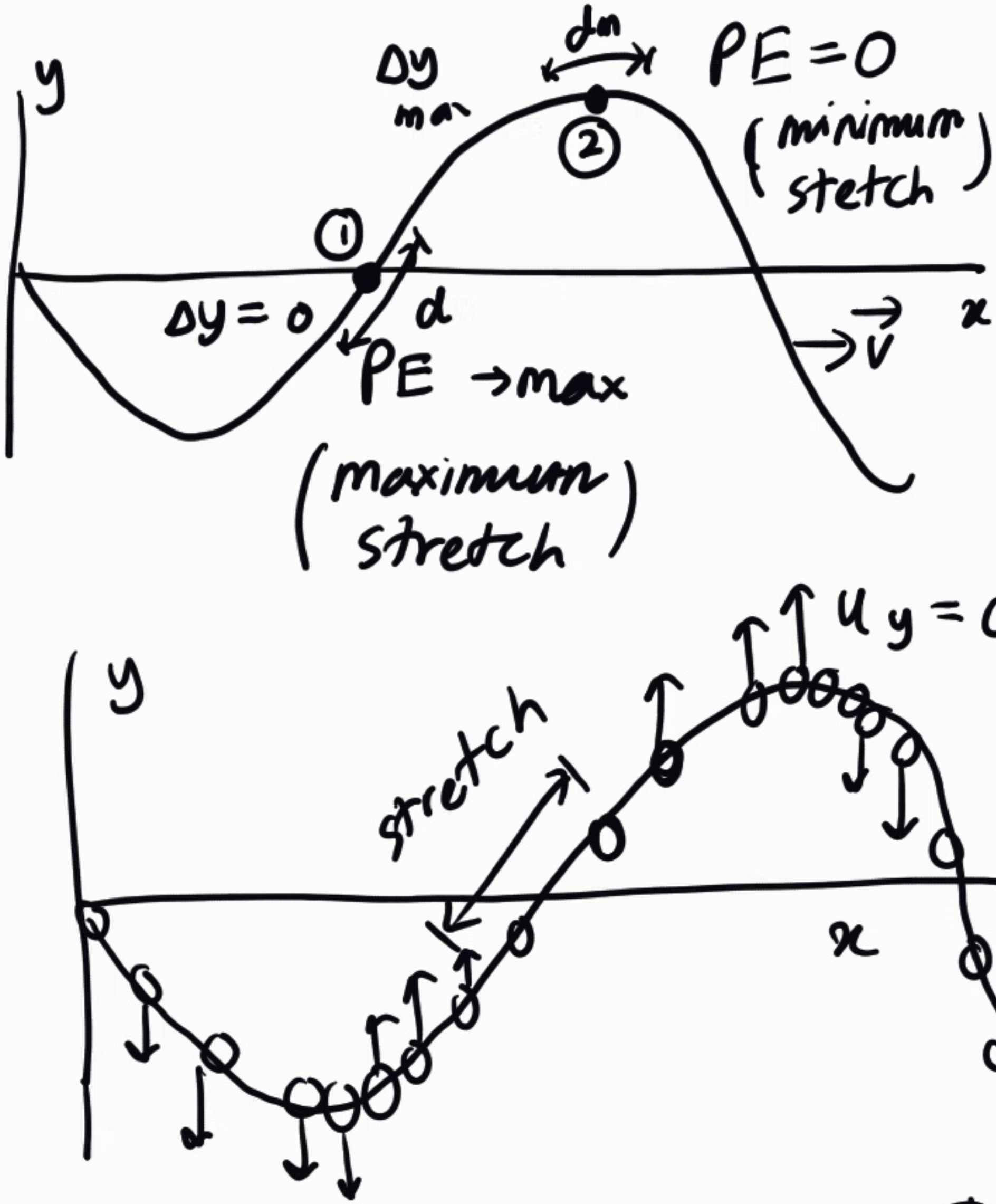
- Travelling wave = transfer of kinetic & potential E.

i, Kinetic Energy (KE)

- mass element, dm oscillating along the y-axis (longitudinal) in a SHM.



ii/ Potential Energy (PE)



- KE of the mass element:

$$dK = \frac{1}{2} dm\, u_y^2$$

- Displacement. Δy

$$y = y_m \sin(kx - \omega t)$$

$$u_y = \frac{\partial y}{\partial t} = -\omega y_m \cdot \cos(kx - \omega t)$$

- mass density (linear density) $= \dfrac{\text{mass element}}{\text{length element}}$

$$\mu = \frac{dm}{dx} \rightarrow \boxed{dm = \mu \cdot dx}$$

$$dK = \frac{1}{2}(dm\, u_y^2)$$

$$= \frac{1}{2}(\mu \cdot dx)\left(\frac{\partial y}{\partial t}\right)^2$$

$$\frac{dK}{dt} = \frac{1}{2}\underbrace{\mu\left(\frac{dx}{dt}\right)}\left(\frac{\partial y}{\partial t}\right)^2$$

v : wave speed

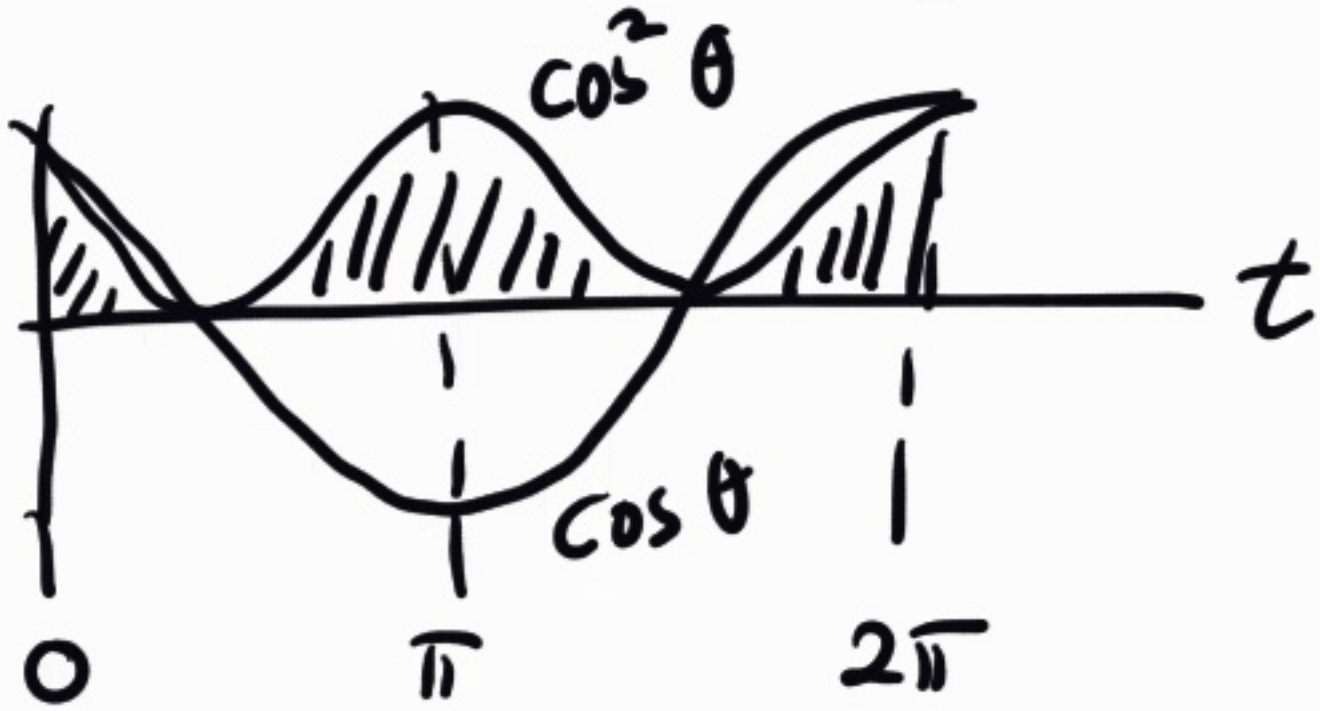
$$= \frac{1}{2}\mu v \omega^2 y_m^2 \cos^2(kx - \omega t)$$

↑ #

Rate of KE transferred

Average rate of KE transferred.

Some Statistics:



$$(\cos^2\theta)_{ave} = \frac{1}{2\pi}\int_0^{2\pi} \cos^2\theta \, d\theta$$

$$= \frac{1}{2}$$

Average Rate of KE Transportation

$$\left[\frac{dK}{dt}\right]_{ave} = \frac{1}{2}\mu v\omega^2 y_m^2 \underbrace{\left[\cos^2(kx-\omega t)\right]_{ave}}_{1/2}$$

$$= \frac{1}{4}\mu v\omega^2 y_m^2 \#$$

- **Elastic Potential Energy** is also carried along the string & it has the same average as KE.

$\therefore$ Average Power: $P_{ave} = 2\left[\frac{dK}{dt}\right]_{ave}$

$$P_{ave} = 2\left[\frac{dK}{dt}\right]_{ave} = \frac{1}{2}\mu v \omega^2 y_m^2$$

Since: $v = \sqrt{\frac{\tau}{\mu}} = \sqrt{\frac{\text{tension}}{\text{linear density}}}$

$$P_{ave} = \frac{1}{2}\mu\sqrt{\frac{\tau}{\mu}}\,\omega^2 y_m^2$$

$$= \frac{1}{2}\sqrt{\tau\mu}\,\omega^2 y^2$$

- Similarly we can determine the average rate of energy transmitted by a sound wave

- Consider a mass element (dm) oscillating back and forth within a thin slice of air with thickness, dx and area, A.

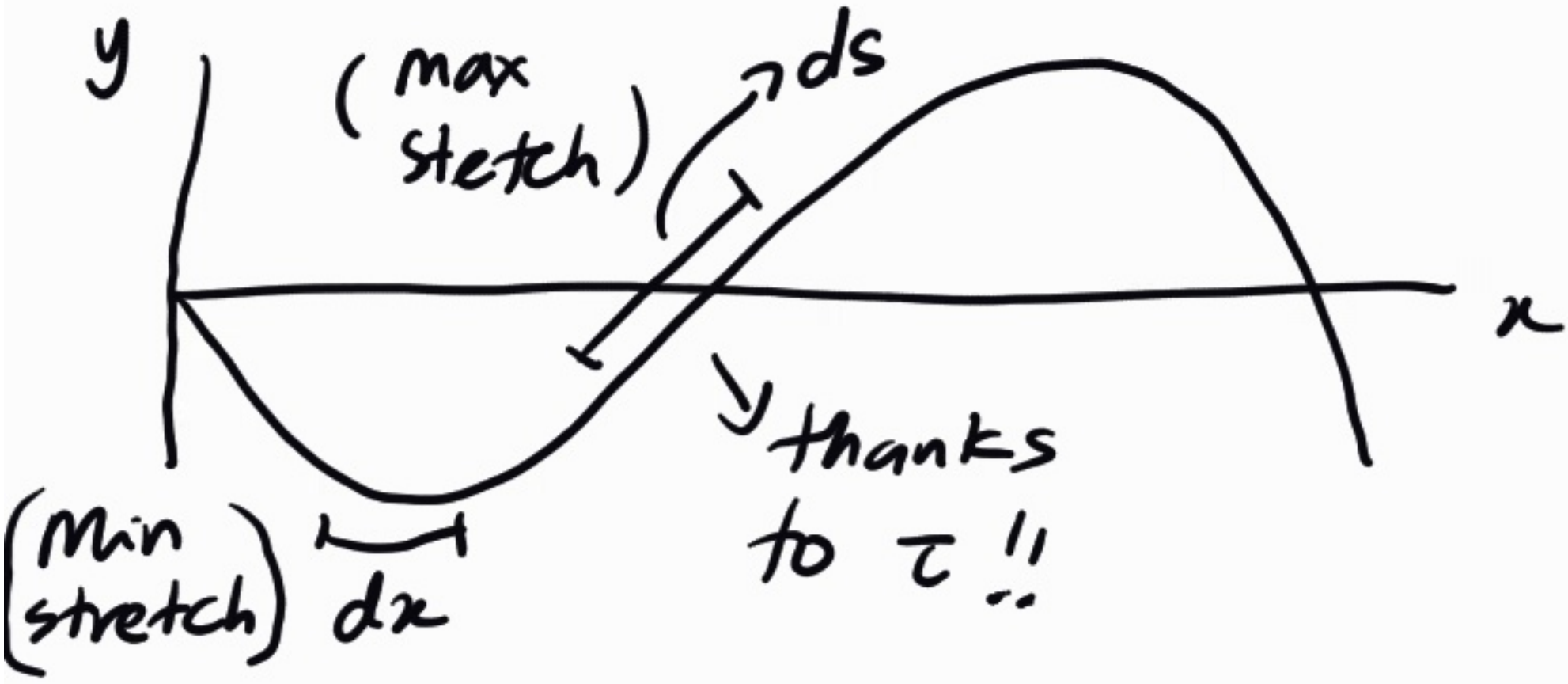
$$dm = \rho dV = \underbrace{\rho A} \cdot dx$$

$$\rho A = \mu \leftarrow \text{unit check}$$

- Power of Sound:

$$\boxed{P_{ave} = \frac{1}{2} \rho A v \omega^2 y_m^2}$$

Elastic Potential Energy

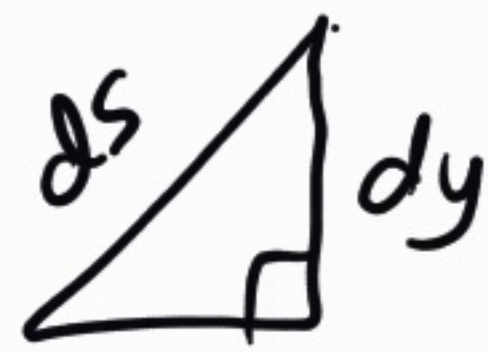


Math Prep

$$ds = \sqrt{dx^2 + dy^2}$$

$$= dx\sqrt{1 + \left[\frac{dy}{dx}\right]^2}$$

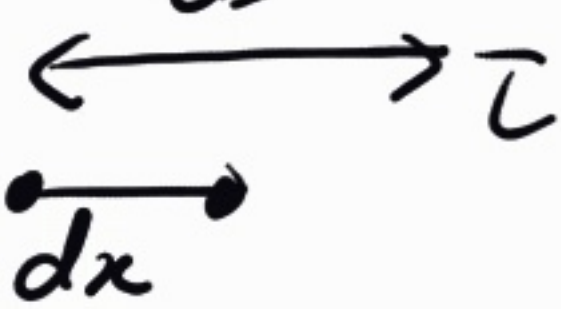
$$= dx\left[1 + \frac{1}{2}\left(\frac{dy}{dx}\right)^2\right]$$



For $x \ll 1$

$$\sqrt{1+x} \approx 1 + \frac{x}{2}$$

→ Let's determine the work done by tension force on a stretched string.



$$dl = ds - dx$$

$$= dx\left[1 + \frac{1}{2}\left(\frac{dy}{dx}\right)^2\right] - dx$$

$$= \frac{1}{2}\left(\frac{dy}{dx}\right)^2 dx$$

$$dW_T = F \cdot dl = T\, dl$$

$$= \frac{T}{2}\left(\frac{dy}{dx}\right)^2 dx = dV$$

elastic potential energy

- Energy transmitted as elastic PE:

$$\frac{dV}{dt} = \frac{\tau}{2}\left(\frac{dy}{dx}\right)^2 \underbrace{\frac{dx}{dt}}_{v\ (\text{wave speed})} = \frac{\tau v}{2}\left(\frac{dy}{dx}\right)^2$$

- Average wave acceleration.

$$y = y_m \sin(kx - \omega t)$$

$$\frac{dy}{dx} = k y_m \cos(kx - \omega t)$$

$$\left(\frac{dv}{dt}\right)_{ave} = \frac{\tau v}{2} k^2 y_m^2 \underbrace{\left[\cos^2(kx-\omega t)\right]^{1/2}}_{\text{cancels}}$$

$$= \frac{1}{4}\mu\omega^2 v y_m^2$$

$$v = \sqrt{\tau/\mu} \Rightarrow \tau = \mu v^2 \ \& \ k^2 = \omega^2/v^2$$ (11)

- Power of the Travelling Wave on A String

$$P = \left(\frac{dK}{dt}\right)_{ave} + \left(\frac{dV}{dt}\right)_{ave}$$

$$= \frac{1}{2}\mu\omega^2 v y_m^2 \quad \#$$